Trash Tracker : IOT Based Garbage Monitoring System Using Raspberry Pi

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***Abstract*— Managing waste efficiently in metropolitan cities is a growing challenge, often leading to overflowing garbage bins, environmental pollution, and increased operational costs. TrashTracker is a smart waste management system that leverages IoT and real-time monitoring to optimize garbage collection. Each bin is fitted with ultrasonic sensors that track fill levels empty, half, or full and automatically notify garbage collectors when bins are nearly full. To make collection more efficient, the system uses the A algorithm* to determine the shortest route to full bins, reducing fuel usage and labor costs. Additionally, Google Maps integration allows real-time tracking of bin locations, ensuring better coordination. By preventing waste overflow and optimizing collection, TrashTracker helps keep cities cleaner, improves public health, and enhances operational efficiency. This project promotes smart infrastructure, environmental sustainability, and healthier urban living.**

***Keywords*—IoT, Smart Waste Management, Garbage Monitoring, Ultrasonic Sensor, Real-Time Monitoring, A* Algorithm, Route Optimization, Google Maps Integration, Waste Level Detection, Environmental Sustainability, Smart Cities, Urban Waste Management.**

I. INTRODUCTION

Garbage collection is an essential part of city infrastructure that has immediate impacts on public health, ecological sustainability, and municipal efficiency. Yet, most cities are still plagued by inefficiencies in garbage collection through outdated conventional processes based on rigid schedules instead of real-time information. This results in garbage being picked up too early, consuming unnecessary resources, or collected too late, spilling bins over and making the environment unhygienic. The result of poor waste management is more fuel usage, higher costs, and pollution of the environment. With increasing urbanization and mounting waste levels, there is a need for a smart data-driven way to improve waste collection and disposal.

The PI Trash Tracker is an Internet of Things trash monitoring system that seeks to transform waste collection by combining sensor technology, cloud computing, and smart routing algorithms. The system functions by loading trash cans with ultrasonic sensors that constantly scan for fill levels and categorize them as empty, half-full, or full. When a bin is full, an automatic message is sent to the municipal waste management department to ensure timely pick-up and avoid waste overflow. In contrast to conventional waste collection systems that stick to fixed routes, the PI Trash Tracker employs the A algorithm* to compute the shortest and most direct route for garbage trucks so that they can pick up only full bins rather than covering pre-defined routes.

To further optimize efficiency, the system is integrated with the Google Maps API, allowing real-time monitoring of bin locations and permitting municipal authorities to see waste distribution throughout the city. The feature enhances coordination, maximizes collection scheduling, and keeps the urban space cleaner. By using IoT and smart routing, the PI Trash Tracker reduces fuel expenditure, labor costs, and carbon emissions, making the waste management system more environmentally friendly. Also, by avoiding overflowing bins and ensuring on-time waste collection, the system maintains cleaner public places, reducing disease risks and overall urban cleanliness.

The use of smart technology in waste management marks an important step in the development of smart cities that leverage analytics and real-time data to improve municipal services. The PI Trash Tracker encourages effective use of resources, minimizes environmental footprint, and supports international sustainability objectives. This paper describes the design, deployment, and performance evaluation of the suggested system with a demonstration of its potential contribution to urban cleanness, optimality in collection of municipal refuse, and further development of intelligent urban infrastructure.

II. LITERATURE SURVEY

This section provides an overview of current research on smart waste management systems with emphasis on IoT-based monitoring, route optimization, and cloud-based data

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analytics. Different research studies have investigated novel approaches to enhancing waste collection efficiency, real-time tracking, and resource optimization. The current literature is classified according to the technologies used, such as sensor-based monitoring, cloud computing, and intelligent routing algorithms, all of which play a role in improving contemporary waste management systems. These researches take into consideration major issues like waste spillover, ineffective collection schedules, and excessive operating expenditures, offering information that guides the design and implementation of the PI Trash Tracker.

A. IoT-Based Waste Monitoring

The application of IoT in waste management has been extensively researched to improve garbage collection effectiveness. In [1], a smart waste monitoring system was presented based on ultrasonic sensors to measure garbage fill levels and categorize bins as empty, half-full, or full. The system utilized wireless connectivity to send real-time information to a centralized platform, enabling authorities to remotely monitor bin statuses. The research proved how IoT-enabled trash cans could streamline collection routes, minimize unnecessary trips, and decrease fuel usage.

In [2], a waste collection system based on IoT was proposed by utilizing RFID and GSM modules for real-time tracking and garbage collection notifications automatically. The suggested system assisted the municipal authorities in managing timely garbage disposal with the advantage of decreasing operation costs. Real-time notification and remote monitoring were emphasized as vital factors for waste overflow prevention and public cleanliness improvement in the research.

B. Route Optimization for Waste Collection

Optimal route planning is necessary to minimize fuel costs and enhance waste collection efficiency. In [3], a shortest path algorithm was used to find the most efficient garbage truck collection route. The work illustrated how dynamic routing can greatly enhance the efficiency of waste collection by lowering total transportation costs and environmental footprint. The research indicated that incorporating machine learning-based predictive analytics would further optimize routes.

Another research [4] proposed the A algorithm* for refuse collection, demonstrating its applicability to optimize routes for garbage trucks using real-time levels of bin fills. The study illustrated that dynamically adapting routes according to bin status would reduce travel time, fuel usage, and labor expenses and, in turn, improve urban waste management. This is the methodology adopted in the PI Trash Tracker, in which the A algorithm* determines the shortest path to be followed in efficient waste collection.C. Cloud-Based Waste Management Systems

Cloud computing is vital for real-time storage, processing, and retrieval of data, hence the key feature of smart waste management. A cloud-based waste monitoring system was proposed in [5], which allows municipal authorities to view real-time bin status, create reports, and optimize collection schedules through a web-based dashboard. The research highlighted the significance of cloud integration in enhancing the scalability, remote access, and the ability to analyze data.

In the same way, in [6], a cloud-based IoT waste management system was proposed to gather and analyze waste information through big data analytics. The study emphasized the importance of predictive analytics in predicting waste accumulation patterns, allowing authorities to organize collection schedules more effectively. The study concluded that cloud-based waste management systems greatly improve data-driven decision-making, minimize manual intervention, and optimize waste collection efficiency.

D. Smart Waste Management Using Machine Learning

Recent breakthroughs in machine learning (ML) and artificial intelligence (AI) have immensely enhanced waste collection systems optimization. In [7], a predictive waste monitoring system based on ML algorithms was designed to forecast garbage accumulation patterns from historical data. The study exhibited how ML models can be utilized to examine seasonal patterns of waste generation and assist municipalities in dynamically modifying collection schedules. Through merging sensor information with AI-based predictions, the study demonstrated a 20% drop in unnecessary pickups, resulting in cost savings and efficient waste disposal.

Another research [8] proposed a deep learning-based image recognition system to sort and distinguish between waste types prior to collection. This system decreased the need for manual sorting and facilitated more effective recycling of waste. The study concluded that AI-driven waste classification could increase recycling rates, reduce landfill use, and assist with environmental sustainability objectives.

F. Wireless Communication for IoT-Based Waste Monitoring

Durable wireless communication is essential for real-time performance of IoT-based intelligent bins. A LoRaWAN-based intelligent waste monitoring system was suggested in [9] to support long-distance, low-power sensor-to-cloud-server communication. It was observed that LoRaWAN networks greatly enhanced the reliability of data transmission in metropolitan areas, where conventional Wi-Fi and GSM networks experienced connectivity problems.

In the same vein, in [10], a 5G-supported IoT waste management system was proposed to enable ultra-fast and low-latency communication between trash cans and city officials. The research showed how 5G networks could be used to improve real-time waste monitoring, data analysis,

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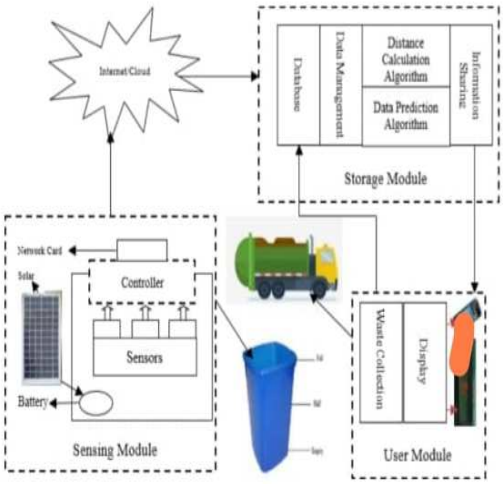
and remote monitoring, which makes it an exciting technology for future smart waste management systems.

III. PROPOSED SYSTEM

The IoT-based Smart Garbage Monitoring System is designed to improve efficiency in waste management through real-time monitoring, data processing, and predictive analytics. There are three modules of the system: Sensing Module, Storage Module, and User Module, each playing an important role in automating waste collection and reducing operation costs.

The Sensing Module is designed to sense the amount of waste in dustbins and provide feedback to a cloud storage system. The module incorporates infrared or ultrasonic sensors to determine the material in the garbage can. Sensor data is processed by a microcontroller as well as observed communication with the cloud. Solar power, augmented with a rechargeable battery, operates the system at an energy-efficient rate. A Wi-Fi, LoRa, or GSM network card enables real-time data transmission to the cloud for remote monitoring.

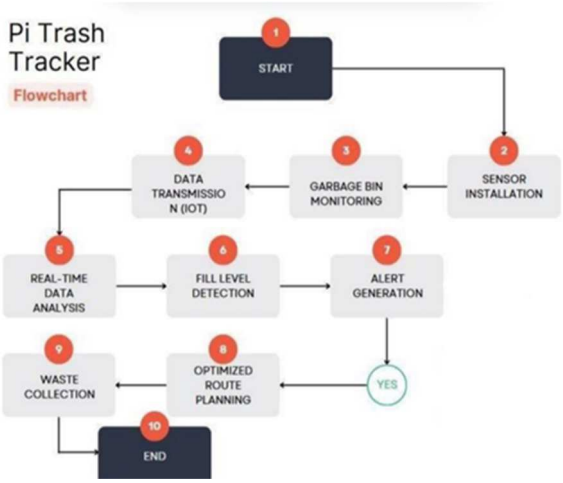


System Architecture

The **Storage Module** is the backend, handling data management and analysis. It contains a database storing real-time and historical garbage level information. Two important algorithms are contained within the module: the Distance Calculation Algorithm, which calculates the fill level from sensor inputs, and the Data Prediction Algorithm, which predicts when bins will be full and schedules collections accordingly. The processed data is also disseminated to waste management authorities to enable informed decision-making.

The **User Module** gives live status updates and helps in waste collection activities. It has a display unit on the garbage bin, which indicates its status as Empty, Half-Full, or Full visually. In addition, the system creates automated reminders and notifications to waste collection teams, allowing them to plan their routes optimally and minimize unnecessary pickups.The system's workflow proceeds through three key steps.

First, the **Sensing Module** gathers current data on garbage levels and sends it to the cloud. Then, the Storage Module analyzes the data, computes bin status, and forecasts fill times. Lastly, the User Module shows the bin status and informs waste collection authorities to maximize collection schedules. the suggested IoT-based Smart Garbage Monitoring System improves waste management through automated garbage level detection, maximized collection schedules, and lowered operational expenses. Through the integration of real-time tracking,cloud-based processing of data, and predictive analytics, the system makes waste disposal on time without overflowing or unnecessary collection.



System Workflow

The Pi Trash Tracker uses a systematic and automated process to effectively track wastage levels and maximize garbage collection. The system starts with the installation of the sensor within assigned garbage bins, where ultrasonic sensors track the level of waste, and gas sensors sense toxic emissions. After the installation, the system continuously tracks the status of the bin, providing real-time data for fill levels and environmental conditions. This data is then transmitted through IoT (Wi-Fi or LoRaWAN) to a central processing unit, e.g., a Raspberry Pi, and then sent to a cloud-based platform for further processing.

During the real-time data analysis stage, the system analyzes the incoming sensor data to identify if a bin has reached its fill level. If the bin is found to be full or if levels of hazardous gases are above safe levels, the system sends an alert and informs municipal authorities or waste collection teams through web or mobile applications. Once the alert has been initiated, the system goes on to optimized route planning, wherein the best collection routes are established based on bin positions and fill levels. This makes waste collection teams go through a smooth process, which results in decreased fuel usage and operating expenses.

When the optimized route plan is received, the waste collection teams go ahead with garbage collection, emptying

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the bins according to the pre-scheduled time. Once the collection is done, the status of the bins is updated in the system, and the process is complete. The system resets for the next monitoring cycle, and the process continues automatically and continuously. Through the incorporation of real-time monitoring, IoT-based data transfer, and smart decision-making, the Pi Trash Tracker improves the efficiency of waste management, minimizes manual intervention, and leads to cleaner, smarter cities. In addition, the system provides detailed reports and analytical information to facilitate municipalities to make informed decisions in upgrading waste management procedures. Continuous optimization not only reduces operational costs but also supports environmental sustainability by facilitating instantaneous and efficient waste disposal.

IV. MODULE DESCRIPTION

1.Sensing Module:

Ultrasonic Sensor (HC-SR04): This is the core of the sensing module. It computes the distance to the level of trash in the bin. It emits ultrasonic pulses and measures the time it takes for the reflected pulse to return. This time is then translated into distance, which is used to compute the fill level. The sensor is attached to the bottom of the lid of the bin**.**

2. Processing Module:

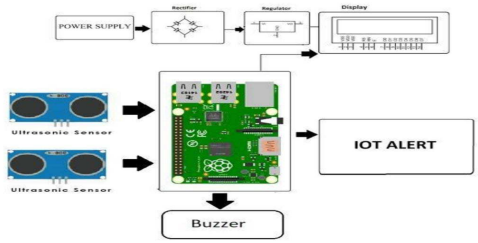
Raspberry Pi:It is the processing module. It receives the distance data from the ultrasonic sensor, processes it, and calculates the fill level percentage. The Raspberry Pi also controls the communication with the cloud or local server. It's chosen for its processing power, connectivity options and the ability to run various software and scripts.

3.Communication Module:

Wi-Fi/Ethernet:The build-in Wi-Fi or Ethernet capability of the raspberry pi is employed to network the system.this allow level data to be transmitted to a cloud service or remote server.this networking plays a central role in remote monitoring.

4. Power Module:

Power Supply: The Raspberry Pi employs a stable power supply in the form of a micro-USB adapter. The ultrasonic sensor will also need to be powered, and this power can be sourced from the Raspberry Pi or an alternative power source. Providing reliable power to the system, especially for remote installations, should be given thought. Solar panels with battery backup, big power banks, or high-efficiency DC-DC converters can be added to keep it running uninterrupted in such a situation. In addition, voltage regulation and overcurrent protection circuits should be integrated to make sure that power variation does not impact system performance. Low-power operation or scheduled sensing periods may also be employed to maximize energy consumption, hence making the system more efficient and sustainable in prolonged deployments.



5.Software Module:

Operating System (Raspbian/Raspberry Pi OS): The Raspberry Pi is powered by a Linux based operating system, including Raspbian (now known as Raspberry Pi OS) which provides the basis on which the required software can be run.

Programming Language (Python): Python is commonly used for programming the system. It is used for controlling the ultrasonic sensor, data processing and handling network communication.

Data Logging and Visualization: The system may further entail the software that log the fill level data over time and the data can be displayed in the form of graphs or charts to depict the waste generation trends.

Web Server/Cloud Platform: A web server (flask, django) can be deployed on raspberry pi or the data can be posted to a cloud platform (aws, google cloud) GPS Module: Including a GPS module assists in monitoring the location of the bins and this is important in determining the best routing for the collection of the bins.

IV. IMPLEMENTATION AND RESULT

The Pi Trash Tracker: IoT-Based Garbage Monitoring System was created using a Raspberry Pi as the processing unit, which was complemented with sensors and cloud platforms for real-time monitoring of waste. The system utilized an ultrasonic sensor (HC-SR04) for the measurement of garbage fill levels, a gas sensor (MQ-135) for toxic gases like methane and ammonia, and a DHT11 temperature and humidity sensor for the measurement of environmental conditions. Wi-Fi (ESP8266 or onboard Raspberry Pi Wi-Fi) was used for data transmission, allowing real-time data transmission to cloud platforms like Firebase, ThingsBoard, or AWS IoT Core using the MQTT protocol. A battery-powered system with optional solar panel integration was also used to ensure continuous operation.

The software module was created through the use of Python scripts designed for the collection, processing, and transmission of sensor data. There was a web and mobile application that facilitated real-time visualization of bin status, alert notifications, and optimized collection routes. Waste levels and gas concentrations were continuously monitored by the system, which activated alarms once bins

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hit specified thresholds. On alarm generation, waste management authorities were alerted to allow timely intervention. The mobile dashboard indicated bin locations while recommending optimized collection routes, thus saving on unnecessary collection trips and boosting operational efficiency.System testing yielded high accuracy results with the ultrasonic sensor having a 90-95% rate of accuracy when it came to fill level measurement. The gas sensor detected hazardous gases and set off alarms when gas levels exceeded safety thresholds. Real-time data transfer was achieved at a latency of less than 2 seconds, thus facilitating fast response times. Prioritization of bins that required action meant that 30% of unnecessary collection runs were saved, hence improved waste management and reduced costs of operations. Waste collection staff reported extensive benefits from the mobile dashboard, given the fact that it enhanced route planning and general efficiency.

the Pi Trash Tracker successfully proved the potential of IoT in streamlining waste management. By reducing chances of waste overflow, optimizing collection, and cutting down operational costs, the system enhances a more intelligent and sustainable waste management process. Future versions can include predictive AI-based analytics to streamline energy efficiency, making the system more scalable. Analytics for waste accumulation prediction and operation through solar power

VI . CONCLUSION

Trash tracker demonstrates the IoT application potential in waste management optimisation in a clear manner. Real-time fill level monitoring with a Raspberry Pi and an ultrasonic sensor enables effective collection routes, prevents overflows, and results in cleaner areas. The project demonstrates an executable and scalable solution for smart cities interested in green waste management methods.

The Trash Tracker initiative adequately proves the sustainability of an IoT waste collection system using a Raspberry Pi and an ultrasonic sensor to offer real-time bin fill levels and optimize collection routes, overflows prevented, and finally a cleaner and greener city.

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